Prop. $\dim(V) < \infty$, B is an orthogonal basis of $(V, \langle , \rangle)$

$T: V \to V$ (对不 orthogonal 的 basis 不成立)

then $T^*_{BB} = (T_{BB})^t$ → transpose

↓

adj. op.

Proof:

Let $T^*_{BB} = (b_{ij})$, $T_{BB} = (a_{ij})$, $B = \{e_1, \cdots, e_n\}$

then $\langle T^*(e_k), e_l \rangle = \langle e_k, T(e_l) \rangle$ ← (def. of T^*)

↓ ↓

$=$ $=$

(共轭)

by

$\overline{b_{lk}} = a_{kl} \iff b_{lk} = \overline{a_{kl}}$

Hence

Self-adjoint Operator

Def.: T is self-adjoint

$\Leftrightarrow T^* = T$

E.g. 1: $V = \mathbb{R}^n$, $\langle , \rangle$ is usual dot product

$T: V \to V$, $T(\underline{x}) = A\underline{x}$

then $T^*: V \to V$, $T^*(\underline{x}) = A^t\underline{x}$

So T is self-adj. $\Leftrightarrow A = A^t \Leftrightarrow A$ is symmetric matrix

E.g. 2: $V = \mathbb{C}^n$, $\langle , \rangle$ is usual dot product

$T: V \to V$, $T(\underline{x}) = A\underline{x}$

then T is self-adj. $\Leftrightarrow \overline{A^t} = A$

(Goal: (2 ways))

Prop. 1: All eigenvalues of self-adj. op. are real

Proof:

Let $\underline{v} \in V$ be any e.vector of $T: V \to V$ with e.value λ and T is self-adj.

Consider $\langle T(\underline{v}), \underline{v} \rangle = \langle T^*(\underline{v}), \underline{v} \rangle$

$= \langle \lambda\underline{v}, \underline{v} \rangle$ $= \langle \underline{v}, T(\underline{v}) \rangle$

$= \overline{\lambda}\langle \underline{v}, \underline{v} \rangle$ $= \langle \underline{v}, \lambda\underline{v} \rangle$

$= \lambda\langle \underline{v}, \underline{v} \rangle$

$\Rightarrow \overline{\lambda} = \lambda \Rightarrow \lambda$ is real

Prop. 2: Let $\underline{v}$, $\underline{w}$ be e.vectors of T ($T(\underline{v}) = \lambda \underline{v}$, $T(\underline{w}) = \mu \underline{w}$) with e.values λ, μ and $\lambda \neq \mu$ and T is self-adj.

Then $\underline{v} \perp \underline{w}$

Proof: (

(★) (important)

Prop. 3: $T: V \to V$ is self-adj.

Suppose $U \leq V$ is T-invariant

$\Downarrow$

Then $U^\perp \leq V$ is T-invariant

Proof:

Let $\forall \underline{w} \in U^\perp$ (WTS: $\forall \underline{w}$, $T(\underline{w}) \in U^\perp$)

For $\underline{u} \in U$, $T(\underline{u}) \in U$

Consider $\langle T(\underline{w}), \underline{u} \rangle$

$= \langle T^*(\underline{w}), \underline{u} \rangle$ ← (★) ← (T is self-adj)

$= \langle \underline{w}, T(\underline{u}) \rangle$

$\in U^\perp$ $\quad$ $\in U$

$= 0$

$\Downarrow$

$T(\underline{w}) \in U^\perp$

Prop 4: T is self-adj.
Then T must have an e.vector ← (Prop 1 with real e.values)

Proof:

Case 1: V is $\mathbb{C}$ inner product space

$\Rightarrow \chi_T(x) = (x-\lambda_1)\cdots(x-\lambda_n)$
where $\forall i, \lambda_i \in \mathbb{C}$

(fundamental theorem of algebra) $\Rightarrow \exists i, \lambda_i \in \mathbb{R}$ and it has an e.vector ②

(?)

T is self-adj. $\Rightarrow \lambda_i \in \mathbb{R}$

Case 2: V is $\mathbb{R}$-vector space
(with a basis $\{\underline{v}_1, \cdots, \underline{v}_n\}$ ②($\mathbb{R}$))
with $V = \mathrm{Span}_{\mathbb{R}}\{\underline{v}_1, \cdots, \underline{v}_n\}$ ①

Consider:

(i) $V_{\mathbb{C}}$: $\mathrm{Span}_{\mathbb{C}}\{\underline{v}_1, \cdots, \underline{v}_n\}$
⇒ ① ($V_{\mathbb{C}}$ is $\mathbb{C}$-vector space)

(ii) $\langle\ ,\ \rangle : V_{\mathbb{C}} \times V_{\mathbb{C}} \to \mathbb{C}$ given by

$$\langle \sum_{i=1}^{n} \alpha_i \underline{v}_i, \sum_{j=1}^{n} \beta_j \underline{v}_j \rangle_{\mathbb{C}} := \sum_{i=1}^{n}\sum_{j=1}^{n} \overline{\alpha}_i \beta_j \langle \underline{v}_i, \underline{v}_j \rangle_{\mathbb{R}},$$

where $\forall i, \alpha_i \in \mathbb{C}$ and $\forall j, \beta_j \in \mathbb{C}$

(Ex.: Check $\langle\ ,\ \rangle_{\mathbb{C}}$ is an inner product on $V_{\mathbb{C}}$

(接上)

(iii) (見附)

Observations:

(1) $T_{\mathbb{C}}: V_{\mathbb{C}} \to V_{\mathbb{C}}$ is self-adj. in $(V_{\mathbb{C}}, \langle , \rangle_{\mathbb{C}})$

(2) $\chi_{T_{\mathbb{C}}}(x) = \chi_T(x)$

E.g.: For (2), take $B = \{\underline{v}_1, \cdots, \underline{v}_n\}$

Then $\chi_{T_{\mathbb{C}}}(x) := \det([T_{\mathbb{C}}]_{BB} - xI)$

with

(見附)

↓

By Case 1:

(見附)

Theorem 1: (RBB Thm. 11.1)

(Generalize: sym. matrix → self-adj. op.)

Proof:

Let $T: V \to V$ be self-adj.

(Prop. 4) $\Rightarrow$ $\exists$ e.vector u of T with

$\|u\| = 1$